

Satellite Communications

ECE 4644 / 5610 Spring 2026

Week #14

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Topics

- Automatic repeat-request (ARQ)
 - stop-and-wait
 - go-back-N
 - continuous transmission
- VSATs
- Star network, Mesh network
- On Board Processing (OBP)
- VSAT system design / Example

Error Detection: ARQ

- ARQ = Automatic Repeat Request
- Used with packet transmission to request retransmission of packets with errors
- CRC (cyclic redundancy check) is used to detect errors
- Typical packet length is 256 to 2048 bits

Error Detection: ARQ

- Let the probability of a bit being in error be p ,
i.e., $p = \text{BER}$
- $p \ll 10^{-3}$ to avoid having frequent bad packets and too many retransmissions
- We need an expression for determining the rates at which 1-bit and multi-bit errors occur within a packet of n bits
- Welcome the binomial probability function...

ARQ

- Probability of k bit errors in a packet of n bits
- Binomial probability function

$$P(k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$\text{where } \binom{n}{k} = \frac{n!}{k! (n - k)!}$$

and $p = \text{BER} = \text{probability of a bit error}$

ARQ Example

- Text is transmitted as ASCII characters
- Seven or eight bit word – 128 or 256 characters
- Single parity bit and seven bit ASCII characters make up an eight bit codeword
- With a single parity bit, how many types of bit error in a codeword can be detected?

ARQ Example

- Single parity bit can detect when there are 1, 3, 5, or 7 errors
- Cannot tell exactly how many or which bits are in error, only that the codeword is corrupt
- Receive system sends out an ARQ to have a flagged codeword re-sent
- If there are 2, 4, 6, or 8 bit errors the word error goes *undetected*

ARQ Example

- Focus on $k = 2, 4, 6$ or 8 bit errors because $k = 1, 3, 5$ or 7 bit errors will be caught and corrected using ARQ

- Example:

Find probability that there are $k = 2, 4, 6$ or 8 bit errors in an eight bit word ($n = 8$)

For two cases:

(1) $\text{BER} = p = 10^{-6}$ and (2) $\text{BER} = p = 10^{-3}$

ARQ Example

- $P(k) = \binom{n}{k} p^k (1 - p)^{n-k}$

$\binom{n}{k}$ is $n C r (8, 2)$

- For two bit errors $k = 2$

$$\binom{8}{2} = 8! / (2! \times 6!)$$

$$= (8 \times 7 \times 6 \dots) / (2 \times 1 \times 6 \times 5 \times \dots)$$

$$= 8 \times 7 / 2 = 28$$

ARQ Example

- (1) Probability of $k = \text{two bit errors}$ in eight bit word ($n = 8$) with $\text{BER} = p = 10^{-6}$

$$P(2) = (8,2) p^2 (1 - p)^{8-2}$$

$$k = 2 \quad (n,k) = 28$$

$$\begin{aligned} P(2) &= 28 p^2 (1 - p)^6 \\ &= 28 \times (10^{-6})^2 \times (1 - 10^{-6})^6 \\ &\approx 28 \times 10^{-12} \end{aligned}$$

Probability of $k = \text{four bit errors}$:

$$P(4) = 70 \times 10^{-24} \dots \text{negligible}$$

ARQ Example

- (2) Probability of $k = \text{two bit errors}$ in eight bit word ($n = 8$) with $\text{BER} = p = 10^{-3}$

$$P(2) = (8,2) p^2 (1 - p)^{8-2}$$

$$k = 2: \quad (n,k) = 28$$

$$\begin{aligned} P(2) &= 28 p^2 (1 - p)^6 \\ &= 28 \times (10^{-3})^2 \times (1 - 10^{-3})^6 \\ &= 28 \times 10^{-6} \times (0.999)^6 = 2.78 \times 10^{-5} \end{aligned}$$

ARQ Example

- Compare with the default of sending seven bit words without parity checking and ARQ
- Find probability of one bit error in seven bit word with **BER = $p = 10^{-3}, 10^{-6}$** :

$$P(1) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$k = 1: \quad \binom{n}{k} = 7$$

$$P(1) = 7 p (1 - p)^6$$

$$\approx 7 \times 10^{-6} \text{ for } p = 10^{-6}$$

$$\approx 7 \times 10^{-3} \text{ for } p = 10^{-3}$$

ARQ Example

- ARQ has reduced the Probability of an undetected Error (PE) roughly by a square factor
- For $\text{BER} = 10^{-3}$:
 - PE is 7×10^{-3} without parity bit and ARQ
 - PE is 2.8×10^{-5} with single parity bit and ARQ
- Similar improvement for $\text{BER} = 10^{-6}$

ARQ Example

- Bit error rate must be low if we want to use long packets
- E.g. $n = 1024$, $k = 1$, $p = 10^{-4}$

$$PE \approx 1024 \times p^1 = 0.1024$$

- One in every 10 packets is in error
- We will lose a lot of time in retransmissions

Text Transmission Example

- *Satellite Communications* (2nd ed.) text has 536 pages
- Each page has about 500 words $\approx$ 3000 characters
- Average word length is five characters
- Space is an ASCII character

Text Transmission Example

- Transmit book over T1 link at 1.544 Mbps with $BER = 10^{-6}$ or $BER = 10^{-3}$?

536 pages x 3000 characters = 1,608,000 characters

- With eight bit words

1,608,000 ch = 12,846,000 bits

- Time to transmit entire text at T1 rate = 8.33 seconds

Text Transmission Example

- Transmit book with no parity / parity for ARQ and BER = 10^{-6} / 10^{-3}
- **Case 1:** No parity, seven bit characters

Probability of character errors for BER = 10^{-6}

$$PE(1) = 7 p (1 - p)^6 \approx 7 \times 10^{-6}$$

1,608,000 characters gives 11 errors in text

Average of one every 47 pages

Text Transmission Example

- **Case 2: Single parity**, eight bit characters with ARQ
- **BER = 10^{-6}**
- Applied to 2-bit errors because 1-bit errors are detected and corrected

$$PE(2) = 28 p^2 (1 - p)^6 \approx 28 \times 10^{-12}$$

- 1,608,000 characters gives ~ 0 errors in text
- We can send 22,000 books before we have an undetected error

Text Transmission Example

- Can we throttle down on the BER requirement?
- **Case 3:** No parity, seven bit characters
- BER = 10^{-3}

$$PE(1) = 7 p (1 - p)^6 \approx 7 \times 10^{-3}$$

gives 11,256 errors in text

Average of 21 typos per page (intolerable)

Text Transmission Example

- **Case 4: Single parity**, eight bit characters, with ARQ
- **BER = 10^{-3}** (applied to 2-bit errors)

$$PE(2) = 28 p^2 (1 - p)^6 \approx 28 \times 10^{-6}$$

Gives 45 undetected errors in text

Average of ~1 typo every 10 pages (big improvement, possibly tolerable)

Text Transmission Example

- With $\text{BER} = 10^{-6}$ and single bit parity + ARQ, we can ensure near perfect transmission
- How often do we need ARQ?
- Probability of detected character errors with $\text{BER} = 10^{-6}$ ($\approx$ probability of 1 bit errors):

$$P(1) = 8 p (1 - p)^6 \approx 8 \times 10^{-6}$$

1,608,000 characters gives 13 errors in transmission of text with 8 bit characters

- So 13 ARQ retransmissions are required

Text Transmission Example

- Penalty for adding parity checking with ARQ is reduced *throughput* = rate at which traffic bits are output
- In this example, if the link were through a GEO satellite, a time of $2 \times .240 \approx 0.5$ sec, would be required for each re-send
- The receipt of the message could then be delayed by as much as $13 \times 0.5 = 6.5$ sec
- Perhaps okay for sending data, less so for voice communications

ARQ on Satellite Links

- ARQ = Automatic repeat request is used with packet transmission to request retransmission of packets with errors
- There are three ARQ techniques:
 - Stop-and-Wait
 - Go-back-N
 - Continuous Transmission

Stop-and-Wait ARQ

- Transmitter sends a packet and waits for
 - ACK (Acknowledged) or
 - NAK (Not Acknowledged)
- When ACK or NAK is received, transmitter sends next packet (ACK) or re-transmits last packet (NAK)
- Transmitter re-sends the packet if no message is received (assumes the packet was lost) once the waiting period has timed out

Stop-and-Wait ARQ

- TCP-IP has 60 ms time-out window (latency)
- Useless for GEO links because of round trip delay of ~ 500 ms
- Throughput is ~ 2 packets / second
- Technique is only useful for terrestrial circuits with short delays
- Note that receipt of every packet is confirmed with an ACK or a NAK

Go back N ARQ

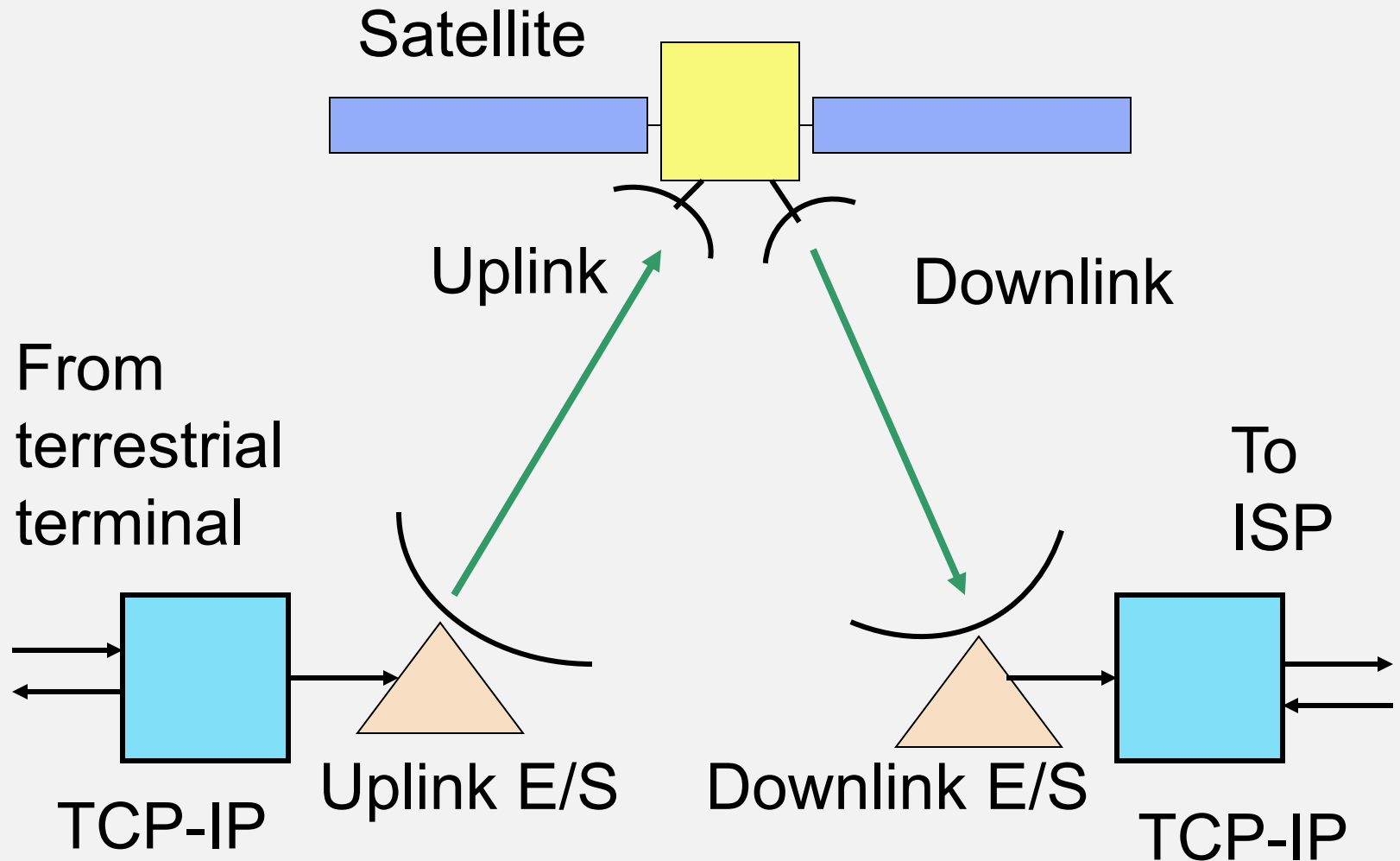
- Transmitter sends numbered packets across GEO satellite link until a NAK is received
- Goes back to the start of the numbered (bad) packet when a numbered NAK arrives
- Retransmits from that point on; for GEO link, 500 ms of packets are discarded at receiver
- Buffering > 500 ms is needed at both ends of the link, adds latency
- Waiting is reduced and throughput is much improved compared to Stop-and-Wait

Continuous Transmission ARQ

- Transmitter sends numbered packets until a NAK is received
- Retransmits numbered (bad) packet only
- Only bad packets are discarded at receiver
- Buffering > 500 ms is needed at ends of link
- Gives maximum throughput of all ARQs
- TCP-IP Internet protocol uses ARQ with continuous transmission

Continuous Transmission

- TCP-IP packets cannot be sent over a GEO satellite link - transmitter times out after 60 ms without an acknowledgement
- *Spoofing* is required at earth station
- Transmitting Earth stations responds promptly to terrestrial sender of TCP-IP packets (with ACK or NAK)
- Terrestrial sender thinks packets went to or from ISP and continues sending packets
- Satellite link uses proprietary protocol to forward packets



Internet access using spoofing of TCP-IP

VSATs (Section 8.5)

- VSAT = Very
Small
Aperture
Terminal
- Antenna (dish) diameter is about 1 m
- Transmit power is 1 - 2 watts
- Data rates are 5 kbps to 4 Mbps

VSATs

A 2.5 m
parabolic dish
antenna for
bidirectional
satellite access

(a somewhat
large VSAT)



VSATs

- Bring telecommunications directly to end user (no intervening terrestrial network)
- Low data rate links use small fixed antennas
- Provides access to remote sites
- Popular for financial transactions, internet access, video, VoIP
- Advanced VSATs have been developed for the NGSO systems

VSATs

- VSATs are a bit larger than DBS-TV dishes (1 m vs. 0.5 m)
- Contrast also with USATs (Ultra Small Aperture Terminal) which are
 - found in satellite telephone handsets
 - omnidirectional



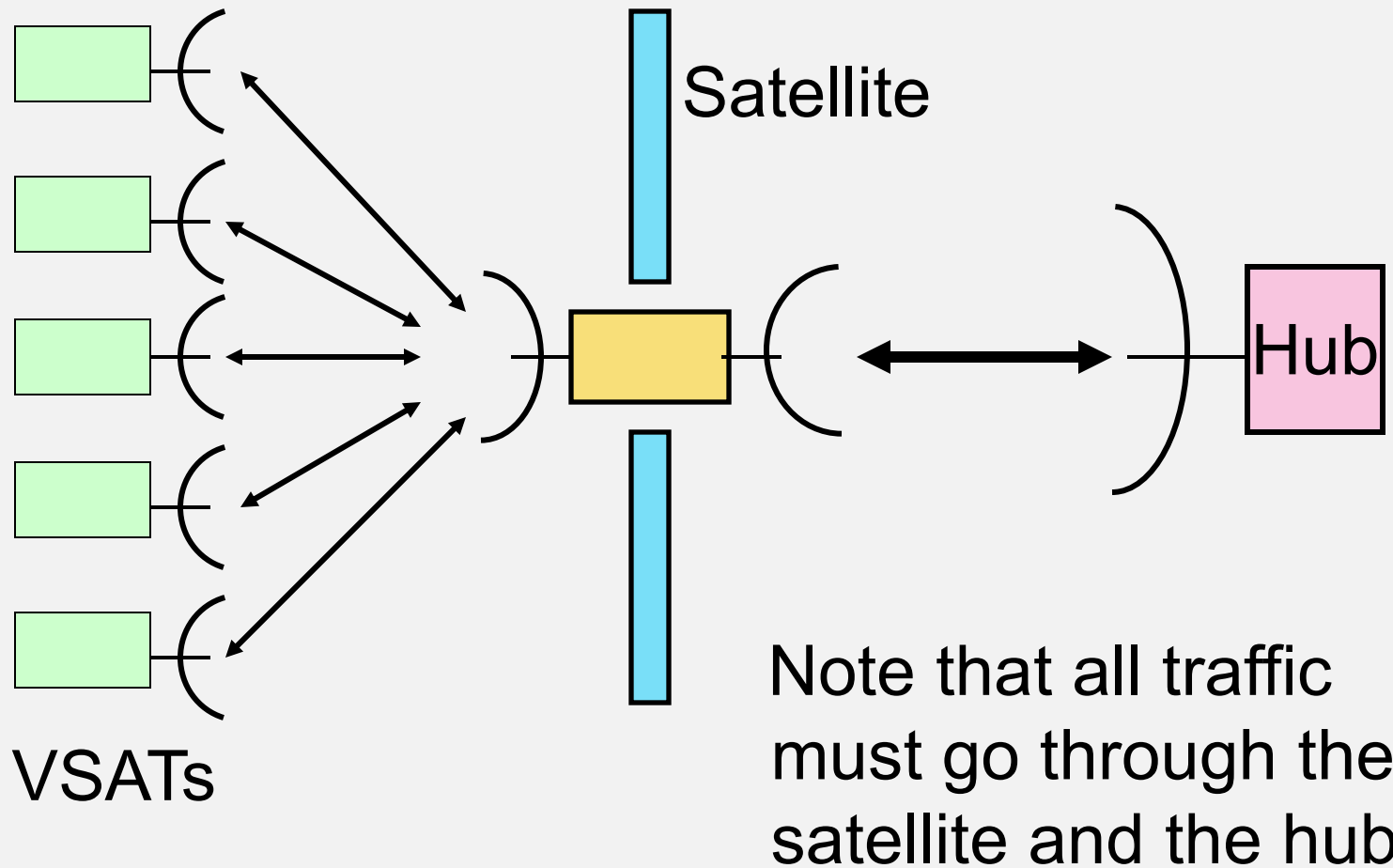
VSATs

- GEO VSAT networks operate in Ku and Ka bands
- Carry data in packets
- For apps where delay is not very important
- Small link margin causes outages in rain
- Central hub is main communication center
- Most networks are star configuration – all data goes through a hub station

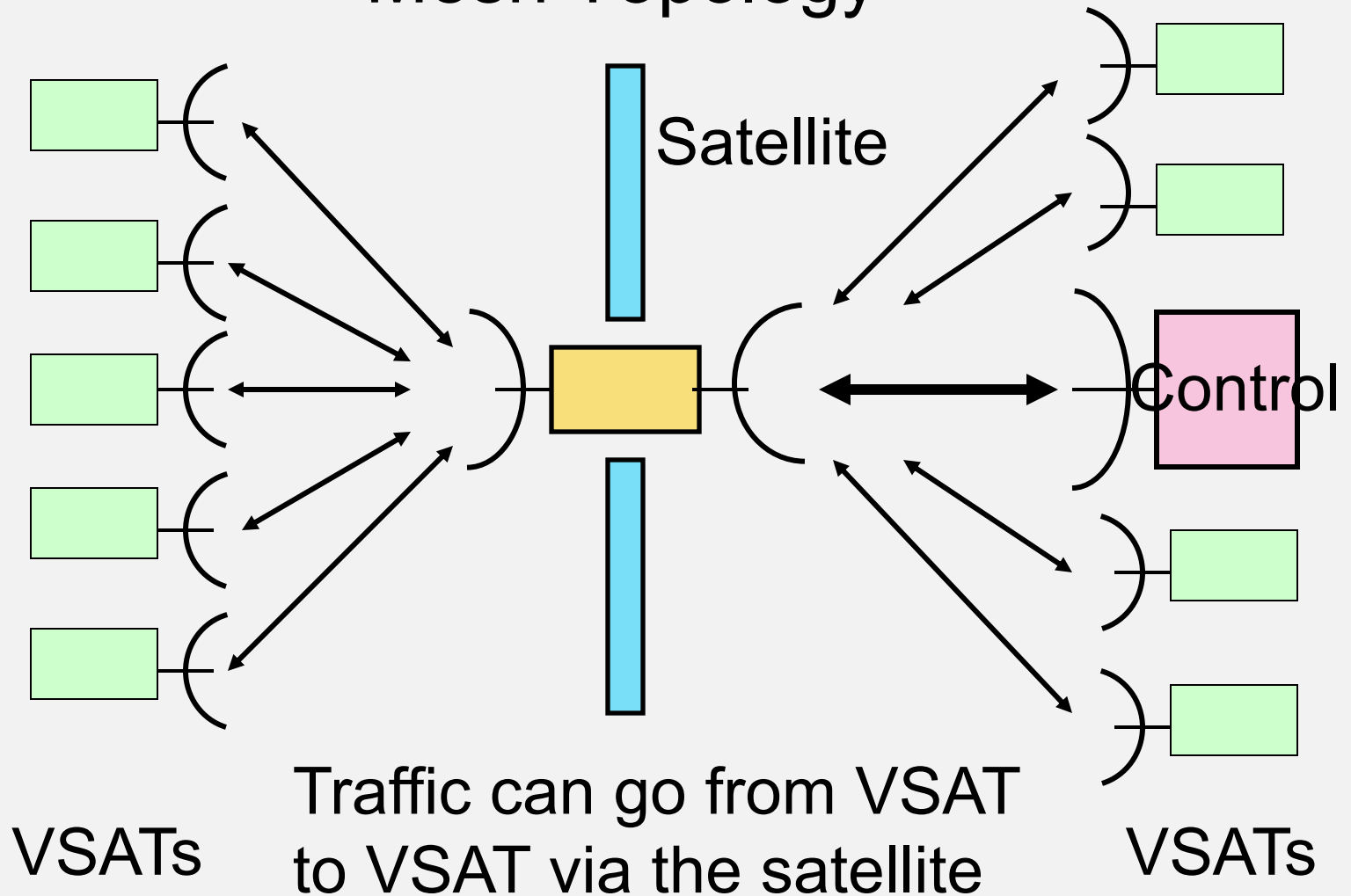
VSAT Implementation for GEO Systems

- There are basically two ways to implement a VSAT Architecture (Section 6.8):
 - 1) *Star*
 - VSATs are linked via a hub
 - 2) *Mesh*
 - VSATs are linked together without going through a large hub

Star Topology



Mesh Topology



Note on G/T ratio for VSAT Stations

- The link equation can be manipulated to show that

$$C/N = k G_r / T_s$$

where k is a constant as far as the receiver is concerned

- The ratio G/T (in units of dB/K) is a measure of receiver quality
- VSAT typically has low G/T on receive and low EIRP on transmit (must be inexpensive)

Advantages of Star Network

- Low EIRP of VSAT is compensated by large G/T of the Hub earth station
- Low G/T of VSAT is compensated by large EIRP of Hub earth station
- Earth station cost is kept low ~ \$10 k
- Monitoring usage is easy with all traffic routed through hub

Disadvantages of Star Network

- VSAT terminals cannot communicate directly with each other; they have to go through the hub
- VSAT to VSAT communications are double-hop
- Double-hop delay might not meet user requirements

Advantages of Mesh network

- Users can communicate directly with each other without being routed via a hub earth station
- VSAT to VSAT communications are single-hop
- GEO Mesh networks have just the usual GEO link single-hop delay

Disadvantages of Mesh Network

- Lower EIRP and G/T now at both ends of a link (compared to Star network gateway station)
- Larger antenna and more transmitted power needed in VSAT terminal = higher cost
- Greater complexity in terms of allocating and tracking usage

NGSO VSAT networks

- New Non-GeoStationary Orbit (NGSO) systems work as VSAT networks
- Operate in Ku, Ka, and V (40 - 75 GHz) bands
- Latency issues are greatly reduced (~ 4 ms one-way)
- Requires sophisticated phased-array antennas to enable links with fast-moving LEO satellites



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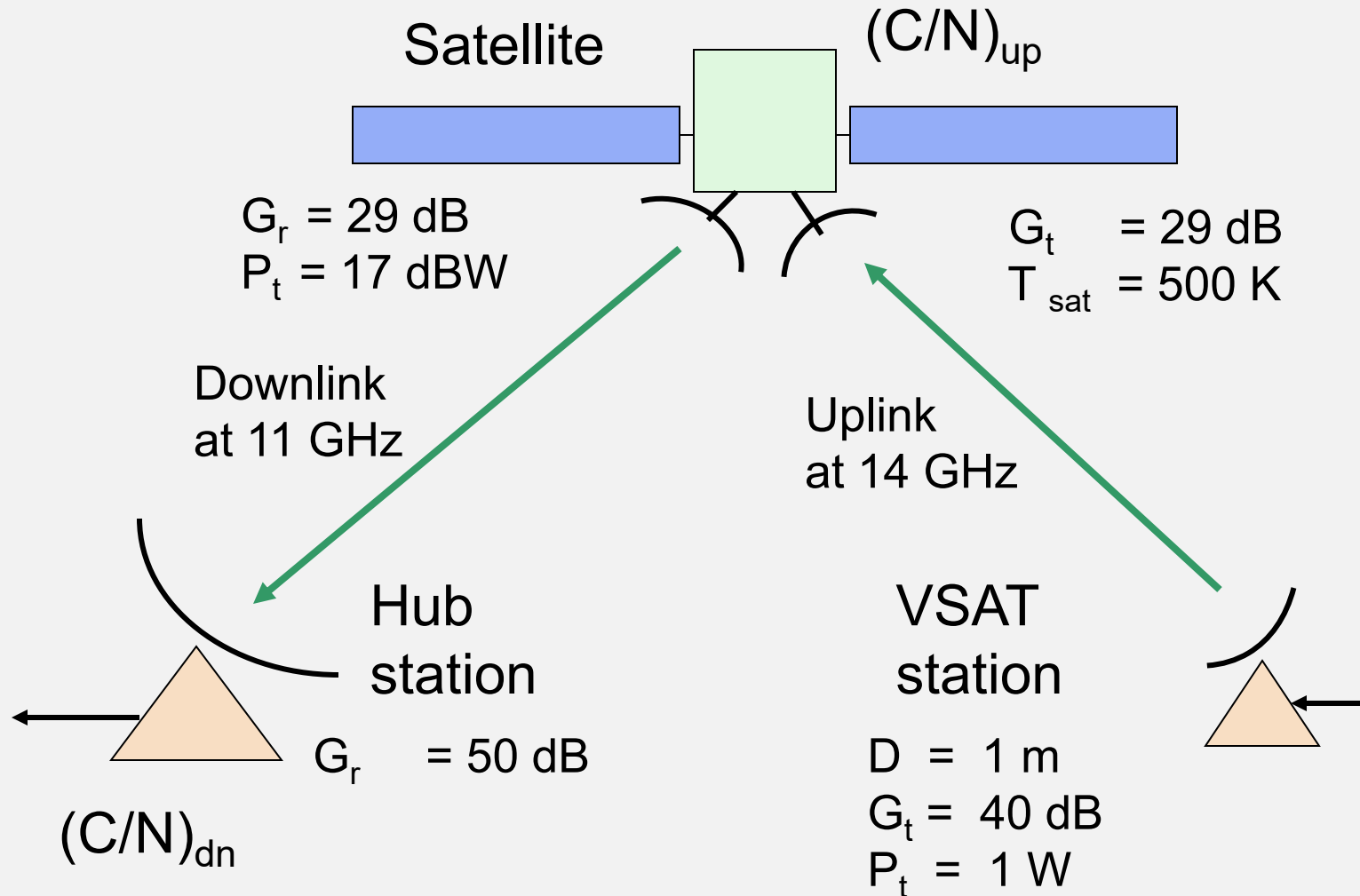
Starlink announced in February, 2022 a new “premium” Tier satellite dish, \$2,500 purchase price and \$500 / month service fee versus \$500 and \$99 / month for non-premium users

VSAT Multiple Access Schemes

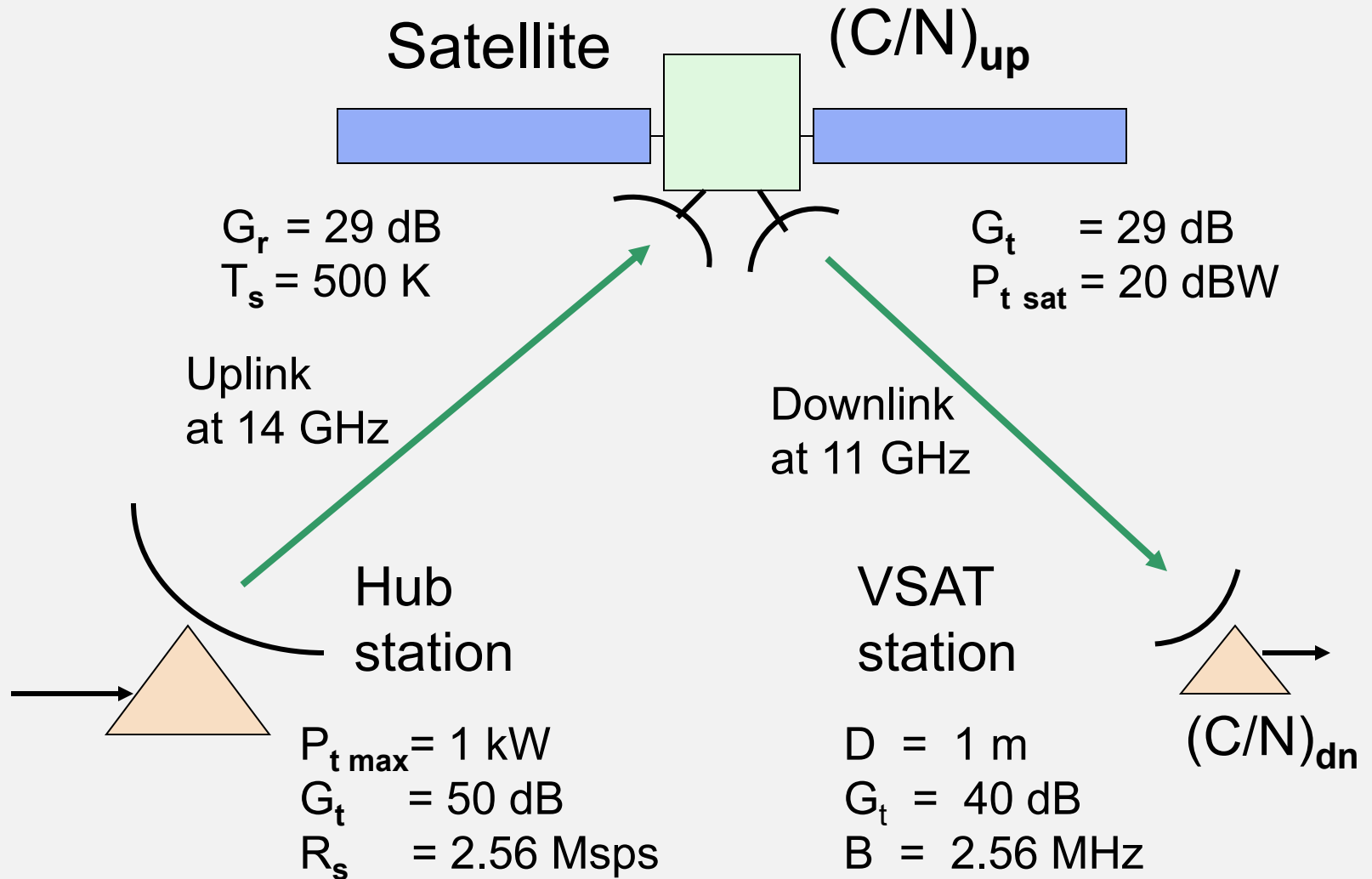
- Choice of FDMA, TDMA, or CDMA for VSAT networks determined by application and economics
- FDMA usually cheapest (lowest power requirement)
- Possible to manage multiple channels with FDMA on both uplinks and downlinks
- Then each VSAT must have tunability to every channel frequency

VSAT Multiple Access Schemes

- With Star mode it is more usual to operate with
 - FDMA from the VSAT on the inbound
 - TDM from the hub on the outbound
- Reminder that we are taking the hub as the reference for describing inbound / outbound
- The VSAT transmits on the inbound and receives on the outbound



Inbound Link: VSAT to Hub – FDMA at 128 ksps



Outbound Link: Hub to VSAT – TDM at 2.56 Msps

Star Network with FDMA-TDM: Inbound

- Note that each route (inbound or outbound) consists of an uplink and downlink
- The inbound is processed via FDMA with each VSAT assigned a transponder channel
- Hub receives many, perhaps hundreds, of downlink VSAT channels

Star Network with FDMA-TDM: Outbound

- For the outbound, the hub assembles the individual VSAT message packets into a single, wideband TDM stream
- Each VSAT receives the stream and peels off its message packets
- Remember the distinction between Time Division Multiplexing (TDM) and TDMA
 - TDM is not a multiple access technique

Star Network with FDMA-TDM: Notes

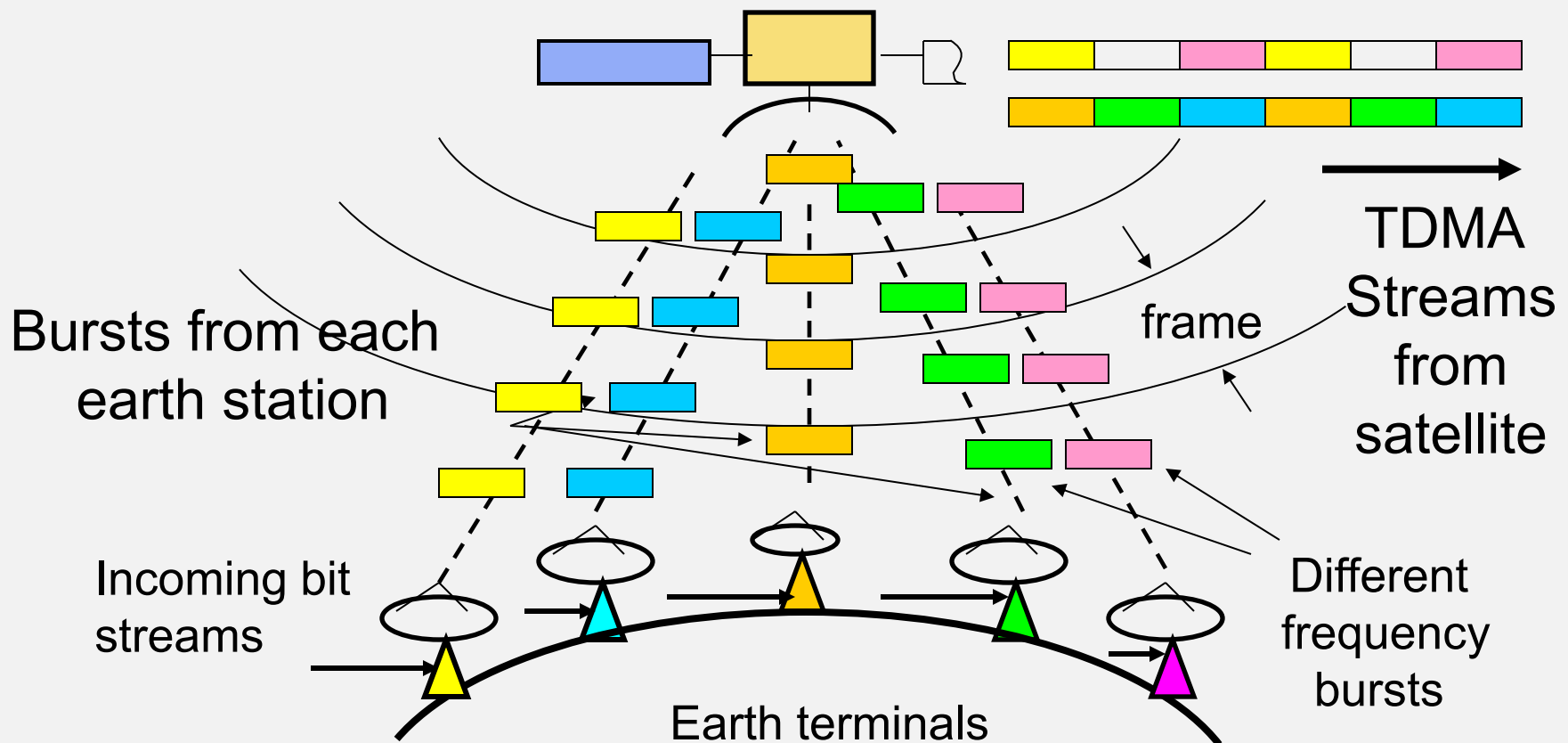
- A single transponder might carry both the inbound narrowband FDMA and the outbound wideband TDM
- Use of FDMA with many channels on the inbound necessitates output standoff on the transponder (3-7 dB)
- Because of cost considerations, the system is likely to operate near margin and suffer occasional outages

Star Network with MF-TDMA

- Popular with Ka band (30/20 GHz) systems
- Reduce FDMA on the inbound by bundling sets of VSAT signals into TDMA
- Thus the hybrid system on the inbound is MF-TDMA
- More bandwidth efficient but requires higher VSAT transmit power, more complexity
- The sets of VSAT signals are still handled by FDMA; outbound from hub is still TDM

MF – TDMA Internet Access

2 FDMA
channels



On-Board Processing (OBP)

- With On-Board Processing (OBP) received signal is demodulated to baseband at the satellite
- Then processed, reassembled, and transmitted onwards
- Compare with bent pipe transponder
- OBP is flexible, in that uplink and downlink are effectively independent
- Adds complexity (and smartness)

Mesh Network: On-Board Processing (OBP)

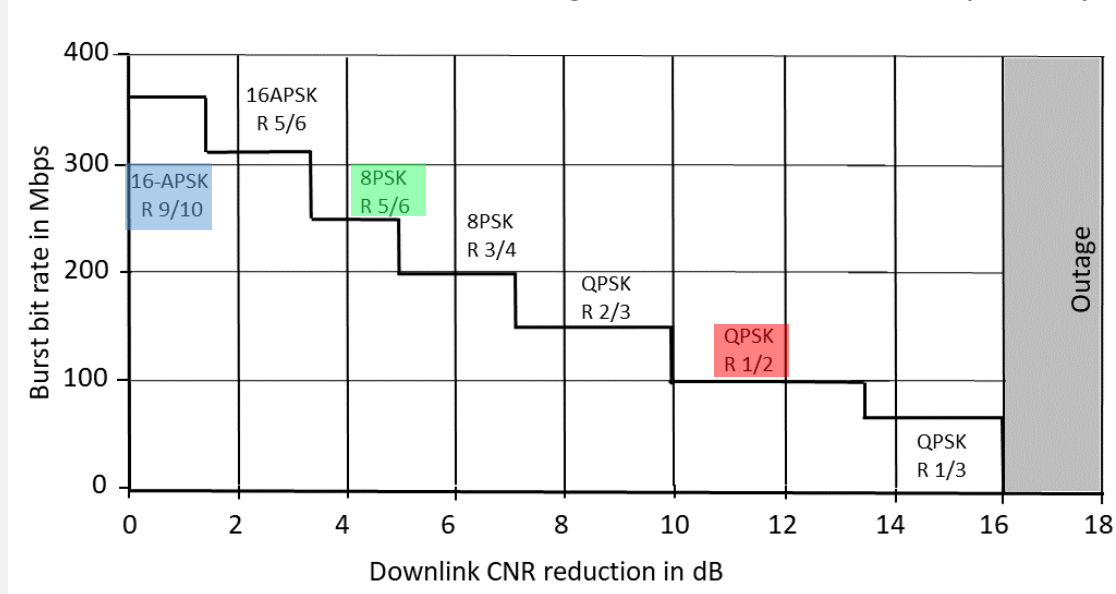
- Can be used to collect FDMA signals from many uplinks:
 - Receive each (narrow-band) uplink signal
 - Process each uplink signal
 - Remove errors, Read addresses
 - Repackage signals into a wideband TDM stream
 - Transmit TDM stream

Mesh Network: On-Board Processing (OBP)

- Advantageous, especially for Ka and V band systems with their sensitivity to rain
- With OBP additional FEC and/or a reduction in modulation order can be applied just to those links affected by rain
- Called *Adaptive Coding and Modulation* (ACM)
- Throughput will decrease on those links
- This is better than having outages

Fig 11.4 page 598 class text

Example of Adaptive Coding and Modulation (ACM)



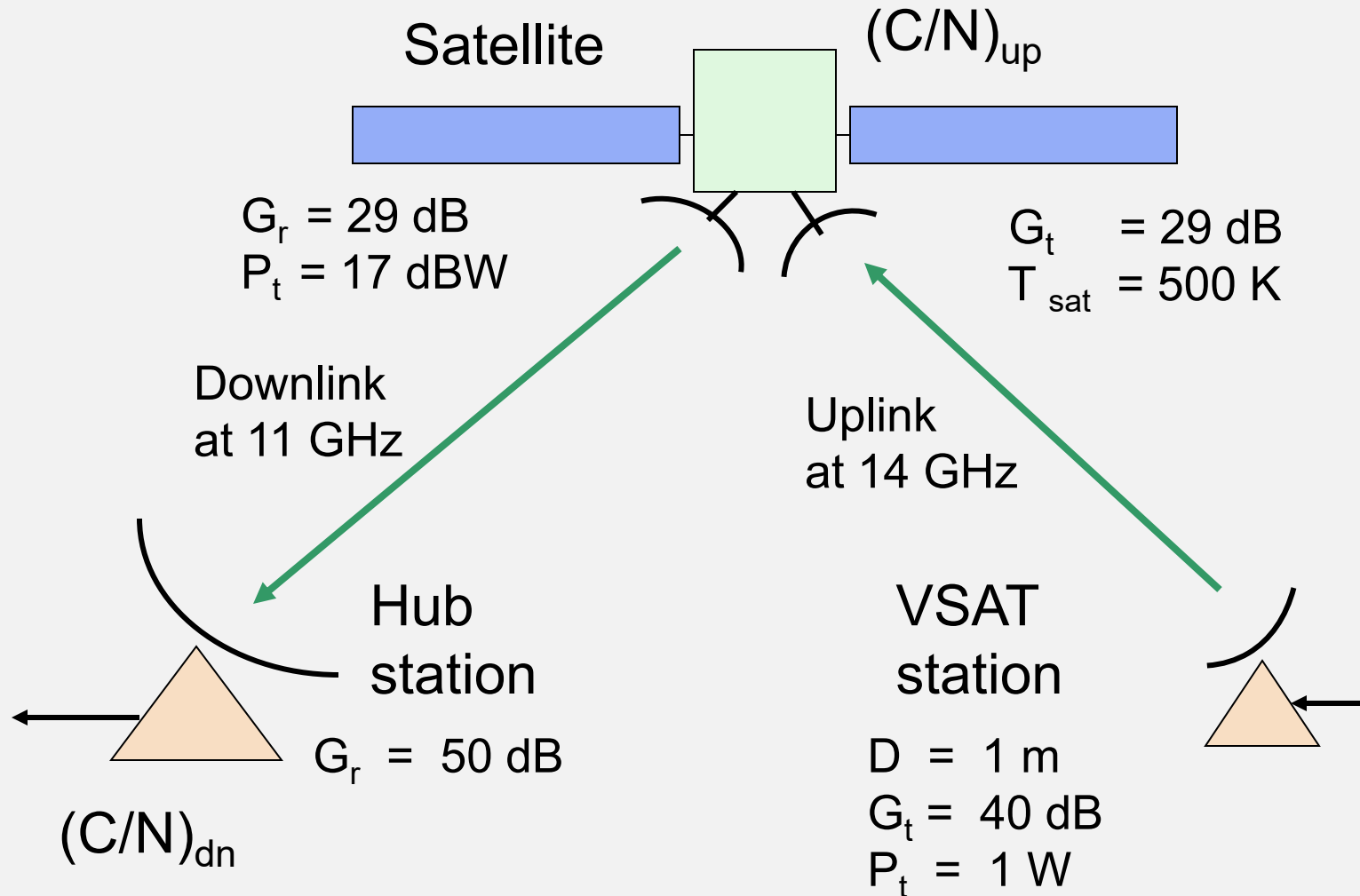
Example of Adaptive Modulation and Coding to combat rain attenuation on downlink

OBP Advantages

- Processing at satellite reduces number of hops
- Detection of signal at satellite enables
 - U/L power control to be exercised
 - On-board routing of traffic (especially with multiple beams)
 - Error detection / correction of uplink signals
- Adaptive coding and modulation can compensate for reduced CNR due to rain

VSAT System Design Example

- Requirements:
 - Design a VSAT Star network using 1 m terminals with 1 W transmitters
 - Transmit and receive 128 kbps data streams in both directions
- Two links must be designed:
 - Inbound link from VSAT to Hub
 - Outbound link from Hub to VSAT



Inbound Link: VSAT to Hub – FDMA at 128 ksps

VSAT System Design - Outbound Link

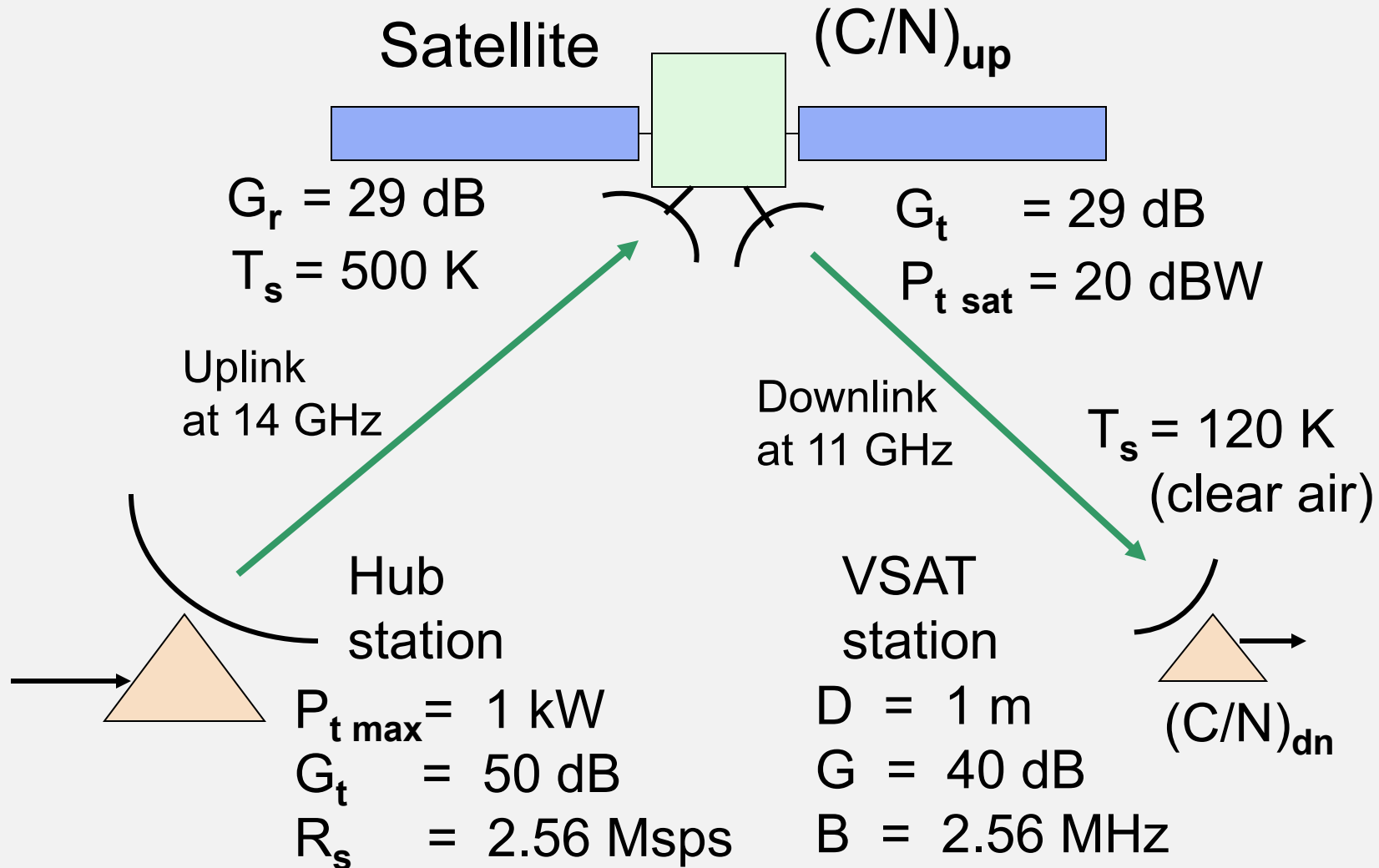
- Large hub station transmits single TDM bit stream
- TDM stream is continuous sequence of packets with addresses
- Each VSAT recovers its own packets
- Specification: $\text{BER} \leq 10^{-6}$ with 4 dB decrease in C/N due to rain
- Link margin is small – outages will occur

VSAT System Design Considerations

- There are four RF links in a two-way VSAT system
- Inbound:
 - uplink from VSAT to satellite transponder
 - downlink from transponder to hub
- Outbound:
 - uplink from hub to satellite transponder
 - downlink from transponder to VSAT

VSAT System Design Considerations

- Each link has its own C/N ratio and requires a separate link budget
- Uplink and downlink C/N ratio are then combined to give an overall $(C/N)_o$
- One for the hub (inbound), one for the VSAT (outbound)
- Bit errors ratios can then be calculated and checked



Outbound Link: Hub to VSAT – TDM at 2.56 Msps

Outbound Link

- We must assume some TDM bit rate to begin design
- Let's assume we have active 40 VSAT stations
- Outbound message bit rate is $40 \times 128 \text{ kbps} = 5.12 \text{ Mbps}$
- Assume no FEC to start

Outbound Link

- Then transmitted bit rate = 5.12 Mbps
- Use QPSK
- Then symbol rate = 2.56 Msps
- Noise bandwidth of VSAT receiver = 2.56 MHz
- Assume $(C/N)_{up}$ is much larger than $(C/N)_{dn}$
- Then $(C/N)_o \approx (C/N)_{dn}$

Outbound Link

- Weak link is the downlink from satellite to VSAT station
- Analyze this link for noise bandwidth $B_n = 2.56 \text{ MHz}$
- Use RRC filters with $\alpha = 0.2$
- Single QPSK symbol stream at 2.56 Msps
- $B_{\text{occ}} = R_s (1+\alpha) = 2.56 \times 1.2 = 3.072 \text{ MHz}$

Outbound Downlink (Satellite – VSAT)

- Use 3 dB back off in transponder

$$P_{\text{out}} = 20 - 3 = 17 \text{ dBW}$$

- Transponder output is 17 dBW = 50 W
- Take 1/5 of this power = 10 W = 10 dBW for outbound downlink
- Build in 5 dB of margin to cover miscellaneous losses and 4 dB loss due to rain (worst case?)

Outbound Downlink Budget (Satellite - VSAT)

- Power budget

P_t satellite	10 dBW
G_t on axis	29 dBW
Edge of satellite beam loss	- 3 dB
Path loss at 11 GHz	- 206 dB
G_r VSAT	40 dB
Link margin	<u>- 5 dB</u>
P_r VSAT	-135 dBW

Outbound Downlink Noise Budget

- Downlink noise budget (VSAT)

k	- 228.6 dBW/K/Hz
T_s (clear air) 120 K	20.8 dBK
B_n 2.56 MHz	<u>64.1 dBHz</u>
N	-143.7 dBW

Outbound Downlink C/N

- $(C/N)_{dn} = -135.0 \text{ dBW} - -143.7 \text{ dBW} = 8.7 \text{ dB}$
- This value includes decrease of 4 dB in $(C/N)_{dn}$ for rain in downlink (recall 5 dB link margin)
- So in clear air we are better by 4 dB and $(C/N)_{dn} = 8.7 \text{ dB} + 4 \text{ dB} = 12.7 \text{ dB}$

Outbound Downlink Analysis

- With the possibility of rain causing 4 dB of loss in downlink $(C/N)_{dn} = 8.7$ dB
- May be too low for comfort
- Switching to $\frac{1}{2}$ rate FEC improves effective C/N by 6 dB (coding gain)
- Then worst case effective $(C/N)_{dn}$
$$(C/N)_{dn} = 8.7 + 6 = 14.7 \text{ dB}$$
- Better than clear air $(C/N)_{dn} = 12.7$ dB but throughput has been halved

Outbound Link Analysis

- Overall $C/N = (C/N)_{dn}$ if $(C/N)_{up} > 30$ dB
- Allow 0.5 dB implementation margin
- Then worst case effective $(C/N)_{dn}$
$$(C/N)_{dn} = 14.7 - 0.5 = 14.2 \text{ dB}$$
- Find BER for QPSK signal for this C/N
- $BER = Q(5.13) = 10^{-7}$ in worst case
- Surpasses minimum $BER = 10^{-6}$

Outbound Downlink Analysis

- Check clear air BER (98% of time)
- Apply same 6 dB of coding gain and 0.5 dB of implementation margin
- $(C/N)_o \text{ effective} = 12.7 + 6 - 0.5 = 18.2 \text{ dB}$
- $\text{BER} = Q(8.13) = 0$
- There are no errors in clear air
- All digital links are designed to give zero errors in normal operation

Outbound Downlink Analysis

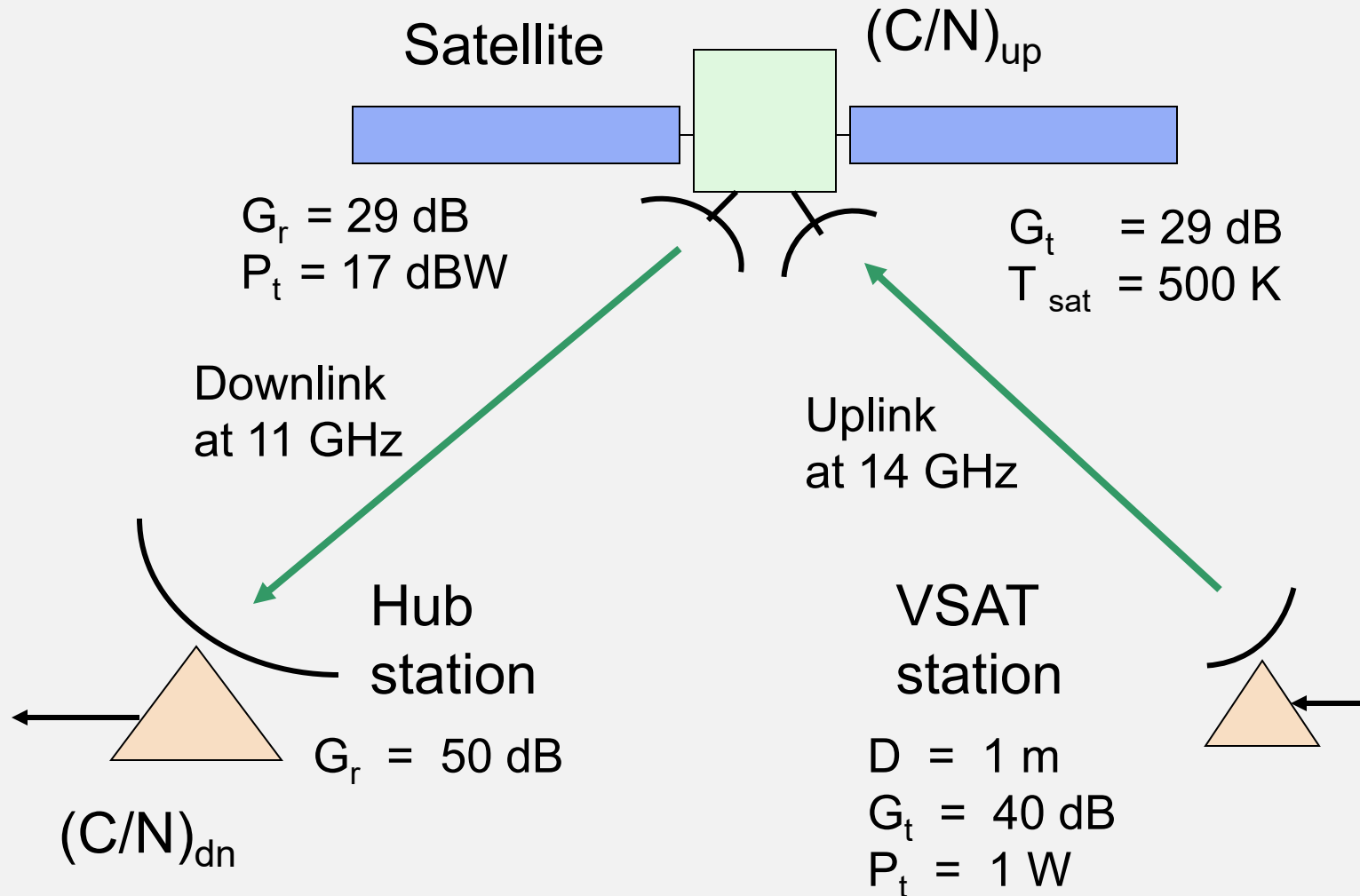
- We are operating acceptably in both clear air $(C/N)_o = 18.2$ dB and heavy rain $(C/N)_o = 14.2$ dB with $\frac{1}{2}$ rate FEC
- Suggests that in clear air we could decrease the FEC rate and increase throughput
- Or switch to a higher modulation scheme, e.g. BPSK to QPSK, QPSK to OPSK
- Or some combination of the two
- That is, apply ACM

Inbound Link – VSAT to Hub

- We assume 40 active VSAT stations
- Inbound data bit rate is 128 kbps
- Use FDMA – best for small stations
- Use RRC filters with $\alpha = 0.2$
- Use QPSK and half rate FEC coding
- Coding gain of ≥ 6 dB is possible

Inbound Link – VSAT to Hub

- Transmitted bit rate = 256 kbps
- Symbol rate = 128 ksps
- Noise bandwidth of hub receiver = 128 kHz
- Assume $(C/N)_{dn}$ is much larger than $(C/N)_{up}$
- Then $(C/N)_o \approx (C/N)_{up}$
- Hub has large, high gain antenna



Inbound Link: VSAT to Hub – FDMA at 128 ksps

Inbound Uplink (VSAT – Satellite)

- Weak link is from VSAT station to satellite
- Analyze this link for noise bandwidth $B_n = 128$ kHz
- Occupied bandwidth is

$$B_{\text{occ}} = 128 \times 1.2 = 153.6 \text{ kHz}$$

- Allow 10 kHz guard band between carriers
- 40 FDMA carriers occupy bandwidth

$$40 \times (153.6 + 10) \text{ kHz} = 6.544 \text{ MHz}$$

Inbound Downlink (Satellite – Hub)

- Use 3 dB back off in transponder

$$P_{\text{out}} = 17 \text{ dBW} = 50 \text{ W}$$

- Take 2/5 of this power for inbound downlink = 20 W
- Each FDMA carrier is allocated $20 \text{ W} / 40 = 0.5 \text{ W}$ at satellite output
- Allow 5 dB margin to cover all losses
- Rain may cause 4 dB reduction in C/N

Inbound Uplink Budget (VSAT to Satellite)

- Uplink power budget

P_t VSAT (1 W)	0 dBW
G_t on axis	40 dB
Path loss at 14 GHz	- 207 dB
G_r satellite	29 dB
Edge of satellite beam loss	- 3 dB
Link margin	<u>- 5 dB</u>
P_r VSAT	-146 dBW

Inbound Uplink Noise Budget

- Uplink noise budget (Satellite)

k	-228.6 dBW/K/Hz
T_s (clear air) 500 K	27.0 dBK
B_n 128 kHz	<u>51.1 dBHz</u>
N	-150.5 dBW

- $(C/N)_{up} = -146.0 \text{ dBW} - -150.5 \text{ dBW}$
 $= 4.5 \text{ dB}$

Inbound Uplink Analysis

- With coding gain of 6 dB for $\frac{1}{2}$ rate FEC and implementation margin of 0.5 dB

$$(C/N)_{\text{eff}} = 10.0 \text{ dB}$$

- $\text{BER} = Q(3.3) = 10^{-4}$
- This design does not meet spec ($\text{BER} \leq 10^{-6}$)
- Recall that this analysis included 4 dB of rain attenuation

Inbound Uplink Analysis

- The BER is too high when rain affects the uplink
- We need $C/N_{\text{eff}} = 13.6 \text{ dB}$ for $\text{BER} = 10^{-6}$
- We must find $\sim 3.6 \text{ dB}$ of additional $(C/N)_{\text{up}}$
- Options:
 - Increase EIRP of VSAT station
 - Increase satellite antenna size
 - Decrease bit rate (and noise power)

Inbound Uplink Analysis

- Or combine options to gain 3.6 dB, e.g.,
split evenly between power and antenna gain
 - Increase P_t to + 1.8 dBW = 1.5 W
 - Increase antenna diameter by 0.9 dB to 1.2 m

Inbound Uplink Analysis

- Worst case $(C/N)_{\text{eff}} = 13.6 \text{ dB}$
 - BER for QPSK = 10^{-6}
- In clear air, $(C/N)_{\text{eff}} = 13.6 + 4.0 = 17.6 \text{ dB}$
 - BER for QPSK = $Q(7.59) = 1.5 \times 10^{-14}$
- There are no errors in clear air
- As expected

Inbound Downlink Analysis

- We should check downlink to hub station
- Each FDMA carrier is allocated 0.5 W at satellite output
- Allow 5 dB margin to cover all losses
- Includes rain attenuation on uplink or downlink = 4 dB
- Calculate link budget and noise budget for downlink

Inbound Downlink Budget (Satellite to Hub)

- Power budget

P_t per VSAT channel	- 3.0 dBW
G_t on axis	29 dB
Edge of satellite beam loss	- 3 dB
Path loss at 11 GHz	- 206 dB
G_r Hub	50 dB
Link margin	<u>- 5 dB</u>
P_r Hub	-138 dBW

Inbound Downlink Noise Budget

- Downlink noise budget (Hub)

$$k \quad -228.6 \text{ dBW/K/Hz}$$

$$T_s \text{ (clear air) } 120 \text{ K} \quad 20.8 \text{ dBK}$$

$$B_n \quad 128 \text{ kHz} \quad \underline{51.1 \text{ dBHz}}$$

$$N \quad -156.7 \text{ dBW}$$

- $(C/N)_{dn} = -138.0 \text{ dBW} - -156.7 \text{ dBW}$
 $= 18.7 \text{ dB} < \text{much higher than } (C/N)_{up}$

Overall Link Analysis

- We have obtained acceptable effective (C/N) and BER on all four links individually
- We should check on $(C/N)_o$ when there is attenuation
- Usually assume one of uplink / downlink is in clear air while the other has 4 dB of attenuation
- Determine percentage of time each link has $BER > 10^{-6}$, report on outage time

Inbound Link Analysis

- Increase in VSAT power (to 3 W) may be needed to ensure $\text{BER} = 10^{-6}$ on inbound link
- We have met all system requirements
- Note that transponders are not used efficiently
- We used 1/5 of outbound transponder power in 3.1 MHz bandwidth (TDM stream)
- We used 2/5 of inbound transponder power in 6.544 MHz bandwidth (FDMA channels)

VSAT System Design Summary

- Set parameter initially to suitable values
- Use single TDM stream outbound, FDMA inbound
- Calculate link budgets and noise budgets for outbound and inbound links

VSAT System Design Summary

- Adjust parameters until requirements are met for worst case
- Check operation in clear air conditions:

$$\text{BER} \Rightarrow 0$$

- Estimate outage times due to rain attenuation for terminals in different geographical locations

